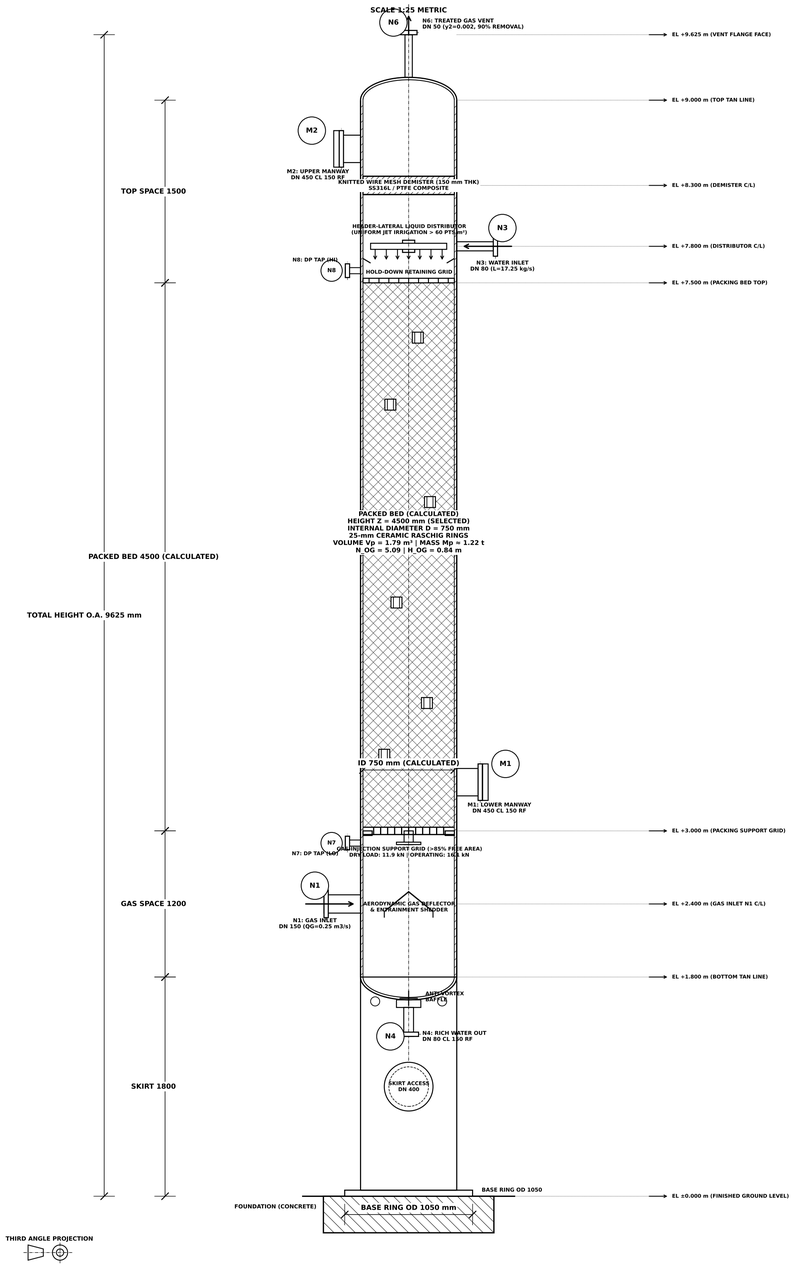
SECTIONAL ELEVATION A-A

SCALE 1:25 METRIC



THIRD ANGLE PROJECTION

EXTERIOR ELEVATION & GA

SCALE 1:25 METRIC

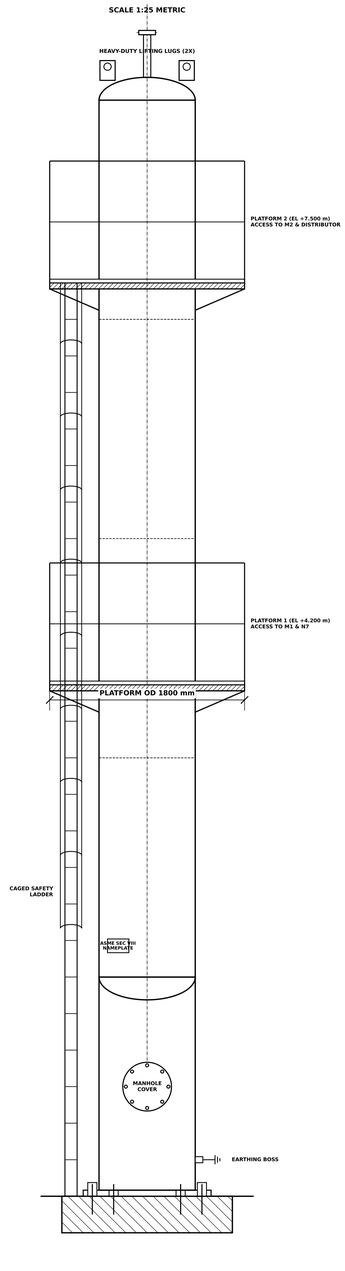


TABLE 1 — PROCESS DESIGN SPECIFICATIONS & SIZING DATA (CALCULATED)

PARAMETER / SPECIFICATION	CALCULATED VALUE	DESIGN BASIS & REMARKS
Equipment Tag & Service	C-101 (Absorber)	Counter-Current Packed Absorption Column
Solute Gas / Carrier Gas	Bromine (Br2) / Air	Dilute Toxic Gas Neutralization / Recovery
Gas Volumetric Flow (QG)	0.25 m³/s (900 m³/h)	Ideal Gas at Operating T and P
Operating Pressure & Temp	803 mmHg (107.06 kPa)	25.0 °C (298.15 K) Ambient Condition
Inlet Br2 Concentration (y1)	2.0 mol% (0.0200)	Raw Off-Gas from Halogenation Header
Outlet Br2 Concentration (y2)	0.2 mol% (0.0020)	90.0% Br2 Recovery / Emission Compliance
Absorbent Solvent / Liquid	Process Water (H2O)	Density = 997 kg/m³, M = 0.018015 kg/mol
Henry's Constant (Hcp)	7.20 x 10⁻⁶ mol/m³Pa	m = pL / (ML * Hcp * P) = 71.8
Equilibrium Line Relationship	y* = 71.8 x	Linear Dilute Henry's Law Regime
Minimum Liquid Rate (Lmin)	683.7 mol/s	x1* = y1 / m = 2.79 x 10⁻⁴
Actual Liquid Rate (L = 1.4Lmin)	957.2 mol/s (17.25 kg/s)	62.10 t/h Scrubbing Circulation Rate
Operating Line Slope (L/G)	90.50	y = 0.002 + 90.5 x
Absorption Factor (A)	0.794	A = m*G / L (Sub-critical Absorption)
Number of Transfer Units (NOG)	5.09 Transfer Units	Colburn Logarithmic Transfer Equation
Selected Packing Type & Size	25-mm Ceramic Raschig	ap=190 m²/m³, ε=0.74, Fp=100 m⁻¹
Packing Bulk Density	680 kg/m³	Dry Bed Mass Mp = 1.22 tonnes
Column Internal Diameter (D)	**750 mm (0.75 m)**	**Calc: 0.73 m at 70% Flooding; Sel: 0.75 m**
Overall Height of Transfer Unit	HOG = 0.84 m	HG = 0.17 m, HL = 0.83 m (Onda Correl.)
Required Packed Height (Z)	**4500 mm (4.50 m)**	Calculated: 4.30 m | Selected Margin: 4.50 m
Active Packing Volume (Vp)	1.79 m³ (1.99 m³ sel)	**Dry Packing Gravity Load: 11.9 kN**
Packing Support Operating Load	16.1 kN (26.4 kN ft)	Dry Packing + Liquid Holdup + Grid Weight

TABLE 2 — PRELIMINARY NOZZLE SCHEDULE (ASME B16.5 CLASS 150 RF)

NOZ	SERVICE DESCRIPTION	SIZE (DN)	RATING	FACING	ELEVATION
N1	Feed Gas Mixture Inlet	DN 150 (6")	CL 150	RF Flange	EL +2.400 m (Gas Space)
N2	Gas Sampling Connection	DN 40 (1.5")	CL 150	RF Flange	EL +2.600 m (Inlet Neck)
N3	Water Absorbent Inlet	DN 80 (3")	CL 150	RF Flange	EL +7.800 m (Distributor)
N4	Rich Liquor Sump Outlet	DN 80 (3")	CL 150	RF Flange	EL -0.300 m (Btm Head Sump)
N5	Column Shell Drain	DN 40 (1.5")	CL 150	RF Flange	EL -0.300 m (Lowest Point)
N6	Treated Overhead Gas Vent	DN 50 (2")	CL 150	RF Flange	EL +9.625 m (Top Apex)
N7	Bed Differential Press (Lo)	DN 50 (2")	CL 150	RF Flange	EL +2.900 m (Below Grid)
N8	Bed Differential Press (Hi)	DN 50 (2")	CL 150	RF Flange	EL +7.600 m (Above Bed)
N9	Gas Temperature Thermowell	DN 40 (1.5")	CL 150	RF Flange	EL +2.200 m (Inlet Gas)
N10	Liquid Level Instrument Taps	DN 50 (2")	CL 150	RF Flange	EL +0.300 / +1.500 m
N11	Demister Flush Spray In	DN 40 (1.5")	CL 150	RF Flange	EL +8.500 m (Above Mesh)
M1	Lower Packing Access Manway	DN 450 (18")	CL 150	RF Flange	EL +3.400 m (Grid Level)
M2	Upper Distributor Manway	DN 450 (18")	CL 150	RF Flange	EL +8.600 m (Top Internals)

ENGINEERING DESIGN CRITERIA & TECHNICAL NOTES

1. DESIGN STATUS CLASSIFICATION (STRICT ENGINEERING COMPLIANCE):

(A) CALCULATED PROCESS PARAMETERS: COLUMN ID 750 mm, PACKED BED HEIGHT 4500 mm, BED VOLUME 1.79 m3,

PACKING MASS 1.22 t, WATER FLOW 17.25 kg/s (62.1 t/h), DRY PACKING LOAD 11.9 kN, N_OG 5.09, H_OG 0.84 m.

(B) PRELIMINARY CAD ASSUMPTIONS: SHELL THICKNESS (8 mm), 2:1 ELLIPSOIDAL HEADS, SKIRT BASE RING OD 1050 mm,

PLATFORMS, CAGED LADDER, NOZZLE PROJECTIONS, AND PIPING LAYOUTS PROPORTIONED FOR GENERAL ARRANGEMENT.

(C) TO BE MECHANICALLY VERIFIED: EXACT SHELL/HEAD THICKNESS PER ASME SEC VIII DIV 1 (UG-27/UG-32) UNDER

INTERNAL PRESSURE (107.06 kPa) + FULL VACUUM CONTINGENCY, SEISMIC/WIND STABILITY PER IS 875 / ASCE 7.

2. CORROSION & METALLURGY SPECIFICATION:

- WET BROMINE AND HYDROBROMIC ACID (Br2 + H2O <-> HBr + HOBr) ARE SEVERELY CORROSIVE TO STANDARD CARBON STEEL.

- COLUMN SHELL MOC: SA-516 GR. 70 WITH SOLID 3.0 mm PVDF (FLUOROPOLYMER) LINER OR TITANIUM GR. 2 CLADDING.

- NOZZLE NECKS & FLANGE WETTED PARTS: PTFE/PVDF LINED OR TITANIUM GR. 2 LINED PER ASTM B363 / B381.

- PACKING: 25-mm CHEMICAL STONEWARE / PORCELAIN CERAMIC RASCHIG RINGS (ZERO CORROSION IN WET HALOGENS).

3. MECHANICAL FABRICATION & TESTING STANDARDS:

- VESSEL FABRICATION CODE: ASME SECTION VIII DIVISION 1 (EDITION 2023), JOINT CATEGORY A & B, FULL RT (UW-11).

- HYDROSTATIC LEAK TEST: 1.5x DESIGN PRESSURE WITH CHLORIDE-FREE DEMINERALIZED WATER; HOLD TIME >= 2 HOURS.

- PNEUMATIC SPARK TESTING OF PVDF LINER AT 15 kV PER NACE SP0188 TO ENSURE 100% PINHOLE-FREE INTEGRITY.

4. CIVIL & FOUNDATION INTERFACE:

- 8x M24 ANCHOR BOLTS (SA-193 B7) EMBEDDED 450 mm IN REINFORCED CONCRETE PEDESTAL (M25 GRADE).

- BASEPLATE TOP GROUTED WITH NON-SHRINK HIGH-STRENGTH EPOXY GROUT (25 mm MINIMUM THICKNESS).

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS EQUIPMENT DESIGN & DRAWING

DRAWING TITLE **PACKED-BED ABSORBER (C-101) — GENERAL ARRANGEMENT & ELEVATION SECTIONAL CAD**	LAB / SUBJECT **PROCESS EQUIPMENT DESIGN & DRAWING**	COURSE / BRANCH **B.E. CHEMICAL ENGINEERING**	SCALE & REVISION **SCALE: 1:25 METRIC / PROPORTIONED | REV: 00**
STUDENT NAME **SOURISH SENAPATI**	EXAM ROLL NUMBER **002310301035**	DRAWING / DOCUMENT NO. **PEDD-PBA-01 (SHEET 01 OF 05)**	DATE **29 / 09 / 2026**
EQUIPMENT IDENTIFICATION & SERVICE **PACKED ABSORPTION COLUMN (C-101) — 750 MM ID x 4500 MM BED**	DESIGN BASIS & METALLURGY / LINING **ASME SEC VIII DIV 1 / SA-516 GR 70 + PVDF LINER / SS316L / CERAMIC**	PROCESS PERFORMANCE & RECOVERY **0.25 M³/S GAS (2 MOL% Br2) | 90% RECOVERY | WATER: 17.25 KG/S**	DESIGN STATUS CLASSIFICATION **CALCULATED: ID 750, Z 4500 | PRELIMINARY CAD: SHELL & NOZZLES**
DIMENSIONS: METRIC (MM / M) | FLANGES: ASME B16.5 CLASS 150 RF | WELDING: ASME SEC IX | THIRD ANGLE PROJECTION			